Mohit Joshi

B.Tech in Computer Science and Engineering | Indian Institute of Information Technology, Una

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Education

Indian Institute of Information Technology, Una — Una, Himachal Pradesh
B.Tech in Computer Science and Engineering — *2025 – 2029*

Nirmala Convent Inter College — Lucknow, Uttar Pradesh
Class XII – 92.5% — *2024*

Technical Skills

Languages: C, C++, SQL, Go, Java, JavaScript

Backend & Databases: Go, PostgreSQL

Tools & Platforms: Git, Vercel, Cursor, Antigravity, Claude Code

Projects

stlgo | *Go, Generics, Data Structures, Algorithms* 2026

- Engineered a high-performance generic data structures and algorithms library for Go, implementing slice-backed Stack, Queue, and Deque containers alongside binary search, sorting, numeric, mathematical, and permutation algorithms.
- Designed containers around contiguous memory layouts and CPU cache locality, using Go generics and type constraints to minimize interface boxing and unnecessary allocations.
- Built unit tests, race-detection workflows, and benchmark suites to validate correctness, memory behavior, and theoretical time-complexity guarantees.

Smart Calculator System | *C++, Shunting-Yard Algorithm, STL* 2025

- Engineered a stack-based C++ expression evaluation engine supporting operator precedence, nested parentheses, floating-point arithmetic, exponentiation, and chained unary operators.
- Implemented context-aware parsing to distinguish unary and binary operators across complex nested mathematical expressions.
- Added robust exception handling for division-by-zero, invalid operators, and undefined power operations involving negative bases.

Leads Tracker Chrome Extension | *HTML, CSS, JavaScript, Chrome API* 2025

- Developed a Chrome extension for saving and managing useful links, supporting custom lead entry and current-tab capture.
- Implemented persistent localStorage-based storage with individual deletion, bulk deletion, and automatic preservation of saved links across browser sessions.
- Improved the extension's user interface and contributed to collaborative development through GitHub issues and pull requests.

Open Source Contributions

Apache Infrastructure Website — Oct. 2025 – Present
Open Source Contributor — *Remote*

- Merged 5 pull requests (#274, #286, #288, #289, #294) addressing frontend, documentation, HTML consistency, and usability improvements.
- Improved HTML/link consistency (#288) and implemented a sticky navigation bar using Bootstrap's `sticky-top` utility (#274).
- Fixed HTML formatting issues and corrected typos and grammatical errors across documentation and services pages (#286, #289).

Achievements

- **4th Runner-Up** – Inter-IIIT Hackathon Udbhav 2025.
- **3rd Position** – PixelForge, IIIT Una.